Chinmoy Bhattacharjee

Junior Professor of Mathematical Stochastics · Universität Hamburg

Personal Details

Place | Date of Birth:	Kolkata, India | 6 December 1990
Nationality:	Indian
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Academic Positions

Sep. 2026–present	**Junior Professor (W1)** at Department of Mathematics, University of Hamburg
Jun. 2026	**Positive interim evaluation** (Habilitationsäquivalent) Extension of Junior Professorship until Aug. 2029
Sep. 2023–Aug. 2026	**Junior Professor (W1)** [first phase] at Department of Mathematics, University of Hamburg
Aug. 2020–Aug. 2023	**Postdoctoral Researcher** at Department of Mathematics, University of Luxembourg
Aug. 2018–Jul. 2020	**Postdoctoral Researcher** at Institut für Mathematische Statistik und Versicherungslehre (IMSV), University of Bern
Jul. 2018–Aug. 2018	**Visiting Scientist** at Indian Statistical Institute, Stat-Math Unit, Kolkata

Education

Aug. 2013–May 2018	**PhD in Applied Mathematics**, Department of Mathematics, University of Southern California Advisor: Prof. Larry Goldstein Thesis: Stein's method and its applications in strong embeddings and Dickman approximations
Aug. 2011–May 2013	**Master of Statistics (M. Stat.)**, Indian Statistical Institute, Kolkata Specialization: Mathematical Statistics and Probability
Aug. 2008–May 2011	**Bachelor of Statistics with Honours (B. Stat. (Hons.))**, Indian Statistical Institute, Kolkata

PREPRINTS

1. **C. Bhattacharjee**, A. Ilienko, B. Jin and I. Molchanov. From Gaussian to stable and back: non-Pareto points in growing dimension. Submitted, 2026+. arXiv: 2611.01096

2. **C. Bhattacharjee**, A. Gusakova, and C. Hirsch. Functional central limit theorems for Laguerre tessellations. Submitted, 2026+. arXiv: 2610.06372

3. **C. Bhattacharjee**. Dickman approximation of randomly weighted sums via Stein's method. Submitted, 2026+. arXiv: 2606.27875

4. **C. Bhattacharjee** and R. O'Clarey. Chaos and superconcentration for Poisson functionals with applications in stochastic geometry. Major revision in *Electronic Journal of Probability*, 2026+. arXiv: 2603.23053

5. **C. Bhattacharjee** and A. Gusakova. Gaussian approximation for extreme points in Laguerre tessellations. Submitted, 2025+. arXiv: 2510.21665

6. **C. Bhattacharjee**, G. Peccati, and D. Yogeshwaran. Spectra of Poisson functionals and applications to continuum percolation. Submitted, 2024+. arXiv: 2407.13502

7. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Multivariate Gaussian approximation for random forest via region-based stabilization. Submitted, 2024+. arXiv: 2403.09960

PEER-REVIEWED PUBLICATIONS

8. **C. Bhattacharjee**, R. Versendaal, and A. Wade. Perimeter length of the convex hull of Brownian motion in the hyperbolic plane. *Journal of Applied Probability*, published online, 2026.

9. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Gaussian and bootstrap approximation for matching-based average treatment effect estimators. *Annals of Statistics*, 54(4), 2164–2190, 2026.

10. **C. Bhattacharjee** and M. Schulte. Dickman approximation of weighted sums of independent random variables in the Kolmogorov distance. *Annals of Applied Probability*, 35(5), 3271–3309, 2025.

11. **C. Bhattacharjee**, I. Molchanov, and R. Turin. Central limit theorem for a birth–growth model with Poisson arrivals and random growth speed. *Advances in Applied Probability*, 56(3), 1004–1032, 2024.

12. **C. Bhattacharjee** and I. Molchanov. Gaussian approximation for sums of region-stabilizing scores. *Electronic Journal of Probability*, 27, paper no. 111, 2022.

13. **C. Bhattacharjee**. Gaussian approximation for rooted edges in a random minimal directed spanning tree. *Random Structures & Algorithms*, 61(3), 462–492, 2022.

14. **C. Bhattacharjee** and M. Schulte. Large degrees in scale-free inhomogeneous random graphs. *Annals of Applied Probability*, 32(1), 696–720, 2022.

15. **C. Bhattacharjee** and I. Molchanov. Convergence to scale-invariant Poisson processes and applications in Dickman approximation. *Electronic Journal of Probability*, 25, paper no. 79, 2020.

16. **C. Bhattacharjee** and L. Goldstein. Dickman approximation in simulation, summations and perpetuities. *Bernoulli*, 25(4A), 2758–2792, 2019.

17. **C. Bhattacharjee** and L. Goldstein. On strong embeddings by Stein's method. *Electronic Journal of Probability*, 21, paper no. 15, 2016.

GRANTS

2024–2027 (36 MONTHS)
Superconcentration in random geometric structures
DFG Priority Program 'Random Geometric Systems' (SPP 2265)
ROLE: Principal Investigator
AMOUNT: 219,966 Euros

2026–2028 (24 MONTHS)
Sharp noise sensitivity & exceptional times in continuum percolation
Programmes for Project-Related Personal Exchange (PPP) [DST & DAAD]
ROLE: co-PI, together with Prof. D. Yogeshwaran (ISI, Bangalore, India)
AMOUNT: 11,308 Euros

INVITED TALKS (SINCE 2022)

DEC. 2026 **Workshop on Stein's Method and Its Statistical Applications**, ESSEC Business School, Cergy, France (upcoming)

AUG. 2026 **European Meeting of Statisticians (EMS)**, invited session *Stein-based inference*, Università della Svizzera italiana, Lugano, Switzerland

JUL. 2026 **Second Workshop on Stein's Method and Networks**, University of Oxford, UK

NOV. 2025 **Mathematics Colloquium**, Institut für Mathematik, Universität Osnabrück, Germany

OCT. 2025 **Workshop on Random Geometry, Graphs and Extremes**, IUC Dubrovnik, Croatia

APR. 2025 **Stochastics Seminar**, Institut for Matematik, Aarhus Universitet, Denmark

OCT. 2024 **Workshop on Random Geometry and Particle Systems**, IMT Nord Europe, Lille, France

AUG. 2024 **Bernoulli–IMS World Congress in Probability and Statistics**, invited session *Stein's method and random graphs*, Ruhr-Universität Bochum, Germany

JUN. 2024 **Workshop on Stein's Method and Networks**, University of Oxford, UK

JUN. 2024 **Workshop on Random Geometric Systems** (SPP 2265), Technische Universität Dortmund, Germany

APR. 2024 **Stochastics Seminar**, Institut für Mathematische Stochastik, Universität Münster, Germany

SEP. 2022 **Mini-Workshop: Limit Theorems for Spatial Random Structures**, Ruhr-Universität Bochum, Germany

JUN. 2022 **Stein's Method: The Golden Anniversary**, Institute for Mathematical Sciences, National University of Singapore

Teaching Experience

University of Hamburg	**Stochastische Prozesse** (current, in German) Winter 2026/27
	Stochastik 1 für Studierende der Informatik (in German) Summer 2025, Summer 2026 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Probabilistic Methods in Combinatorics Summer 2025, Summer 2026
	Stochastik 2 für Studierende der Informatik (in German) Winter 2025/26 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Percolation Theory Winter 2025/26
	Introduction to Random Graphs Master in Mathematics; Winter 2023/24, Winter 2024/25
	Introduction to Stein's Method Master in Mathematics; Summer 2024
University of Luxembourg	**Probability and Statistics II** BMATH fourth semester; Spring 2023, Spring 2022
	Refresher in Probability Master of Data Science first semester; Fall 2022, Fall 2021
	Probabilistic Methods in Combinatorics BMATH fifth semester; Fall 2021, Fall 2020
University of Bern	**Introduction to Random Graphs** MMATH second semester; Spring 2019
University of Southern California (Assistance)	**Math 125 - Calculus I**; Fall 2017 **Math 308 - Statistical Inference and Data Analysis II**; Spring 2016 **Math 307 - Statistical Inference and Data Analysis I**; Fall 2015 **Math 226 - Calculus III**; Fall 2014, Spring 2015

Supervision

PhD.	Rowan O'Clarey (Oct. 2024 – Sep. 2027 expected) Funded by DFG grant no. 531540467 Topic: Superconcentration in random geometric structures
Other Supervision	Amélie Boussemart (Master's thesis) Oliver Gertke (Master's thesis) Luzie Grutke (Bachelor's thesis, ongoing) Maika Lu Beecken (Bachelor's thesis)

Organization & Academic Service

Workshop Organization:

MAR. 2025 **Session Co-chair:** *Spatial stochastics, disordered media, and complex networks*, in the *German Probability and Statistics Days*, Dresden, Germany.
Co-chaired with B. Jahnel (TU Braunschweig/WIAS Berlin)

SEP. 2024 **Workshop Co-organizer:** *Workshop on Stein's method and stochastic geometry*, Hamburg, Germany.
Co-organized with M. Schulte (TUHH, Hamburg)

Peer Reviewing:

JOURNALS Annals of Statistics (AoS), Applied Probability Journals, Bernoulli, Electronic Journal of Probability (EJP), Extremes, Journal of the American Statistical Association (JASA), La Matematica, Methodology and Computing in Applied Probability (MCAP), Probability Theory & Related Fields (PTRF), Random Structures & Algorithms (RSA), Sankhyā A, Statistics and Computing, Statistics & Probability Letters, Stochastic Processes & their Applications (SPA)

Thesis Examination:

2026–2027 **Co-examiner** for a doctoral thesis, Technische Universität Hamburg
2024–present **Co-examiner** for Master's theses, Universität Hamburg (two to date)

Language Skills

Bengali (mother tongue), English (C2), German (B2), French (A1), Hindi (C1)